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The Effects of a Weight Belt on Trunk and Leg Muscle Activity and Joint Kinematics During the Squat Exercise

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ABSTRACT

Fourteen healthy men participated in a study designed to examine the effects of weight-belt use on trunk- and leg-muscle myoelectric activity (EMG) and joint kinematics during the squat exercise. Each subject performed the parallel back squat exercise at a self-selected speed according to his own technique with 90% of his 1RM both without a weight belt (NWB) and with a weight belt (WB). Myoelectric activity of the right vastus lateralis, biceps femoris, adductor magnus, gluteus maximus, and erector spinae was recorded using surface electrodes. Subjects were videotaped from a sagittal plane view while standing on a force plate. WB trials were completed significantly faster ($p < 0.05$) than NWB trials over the entire movement and in both the downward phase (DP) and upward phase (UP). No significant differences in EMG were detected between conditions for any of the muscle groups or for any joint angular kinematic variables during either phase of the lift. The total distance traveled by the barbell both anteriorly and vertically was significantly greater ($p < 0.01$) in the WB condition than the NWB condition. The velocity of the barbell was significantly greater ($p < 0.01$) both vertically and horizontally during both the DP and UP in the WB condition as compared with the NWB condition. These data suggest that the use of a weight belt during the squat exercise may affect the path of the barbell and speed of the lift without altering myoelectric activity. This suggests that the use of a weight belt may improve a lifter's explosive power by increasing the speed of the movement without compromising the joint range of motion or overall lifting technique.

Key Words: biomechanics, EMG, powerlifting

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Introduction

Wearing abdominal weight belts while performing heavy lifts is common practice among both recreational weight trainers and competitive powerlifters. Also, both weight trainers and manual material handlers claim perceivable benefits from wearing a weight belt during heavy lifting activities (5). The utility of belts in enhancing performance and reducing the risk of injuries during heavy lifts is still the topic of much debate (4). However, studies have shown little or no change in muscle activity during movements performed with a weight belt (1, 4). Research examining mean myoelectric activity (mEMG) activity of trunk and leg muscles during the squat exercise conducted with a weight belt reported small changes in the mEMG of leg muscles with moderate loads (3) and no changes in the mEMG of trunk muscle with heavy loads (2). Little information exists in the literature describing the effect of weight-belt use on the muscle activity and joint kinematics during single, near-maximal lifts.

The purpose of this study was to investigate the effects of a weight belt on low back and leg muscle activity and lifting kinematics during the performance of the squat exercise at near maximal loads, and to describe the observed differences in lifter technique.

Methods

Subjects

Fourteen healthy men who were currently involved in weight-training programs served as volunteer subjects. Subject physical characteristics were (mean ± *SD*) the following: age, 28.5 ± 3.3 years; height, 173.1 ± 9.5 cm; weight, 87.6 ± 10.6 kg. Each subject completed a health history and exercise questionnaire and was screened for a history of back injury or chronic back pain. Any subject who reported having a history of back injury or pain was excluded from the study. Each subject completed a university-approved informed consent form prior to participation.

Experimental Setup

Subjects lifted Olympic-style barbells and weight plates while standing in a standard 1.25 × 0.75 × 2-m powerlifting cage. The weight belt used was of heavy construction and composed of 3 layers of leather 8 mm thick and 100 mm wide along its entire length.

All trials were videotaped at 60 Hz with a shuttered Sony WV-D5100 video camera positioned orthogonal to the sagittal plane of the subject at a distance of 5 m and at a height of 1.25 m. Reflective joint markers were adhered to the subjects' iliac crest, greater trochanter, head of the fibula, lateral malleolus, head of the fifth metatarsal, and on the end of the barbell. When the belt was worn, the pelvic marker was placed on the weight belt at the closest approximation of the iliac crest. Subjects wore only shorts and athletic shoes during testing. A video playback system consisting of a Sony PVM-1341 video monitor and a Panasonic AG-2400 video cassette recorder interfaced with a computer equipped with digitizing software (Peak Performance Technologies, Inc., Englewood, CO) was used to manually digitize each of the trials.

To monitor myoelectric activity (EMG), a pair of 8-mm Ag/AgCl pregelled disk surface electrodes was placed over the erector spinae (ES) at the level of L3-L4 approximately 5 cm from the midline of the spine and over the muscle bellies of the right vastus lateralis (VL), biceps femoris (BF), adductor magnus (AM), and gluteus maximus (GM). Electrodes in each pair were positioned 2.5 cm apart and in parallel with the underlying muscle fibers. The raw EMG signals were amplified ×5, sampled at 1,000 Hz per channel, bandpass filtered between 10 and 350 Hz, and processed by a 12-bit analog-to-digital (A/D) converter. A Konigsberg TR8 computer system received the amplified signal for processing and storage. EMG data analysis was conducted using a Labview-based software package (VideoSync, Los Angeles).

All trials were performed on a Kistler force platform positioned at the center of the powerlifting cage. Anteroposterior, mediolateral, and vertical ground reaction forces were collected using a Konigsberg TR8 computer via an A/D converter at a sampling frequency of 1,000 Hz.

Procedure

At least 3 days prior to data collection, the 1 repetition maximum (1 RM) effort of the squat exercise was determined for each subject by having the subject lift increasingly heavier loads and self-determine the maximum amount of weight that could be lifted safely. Since a 1RM is typically of greater magnitude when determined with a weight belt than when determined without a weight belt, the 1RM was determined without the use of a weight belt to ensure that each repetition could be completed in each condition.

On the day of data collection, subjects were allowed to stretch and warm up with lighter weights prior to testing. Each subject was instructed to squat according to his own technique to a depth at which his thighs were parallel to the ground (Figure 1). No attempt was made to control for speed since this may alter the lifting technique and would be inadvisable during a lift conducted at near-maximal loads. Each subject then lifted 90% of his 1RM weight (determined prior to data collection) once while wearing a weight belt (WB) and once while not wearing a weight belt (NWB). Subjects were randomly assigned to lift first with the belt or first without the belt. To ensure safety, trials for all subjects were conducted within the powerlifting cage and in the presence of an experienced spotter. All data were collected for each subject on the same day, allowing for adequate rest (at least 5 minutes) between each lift.

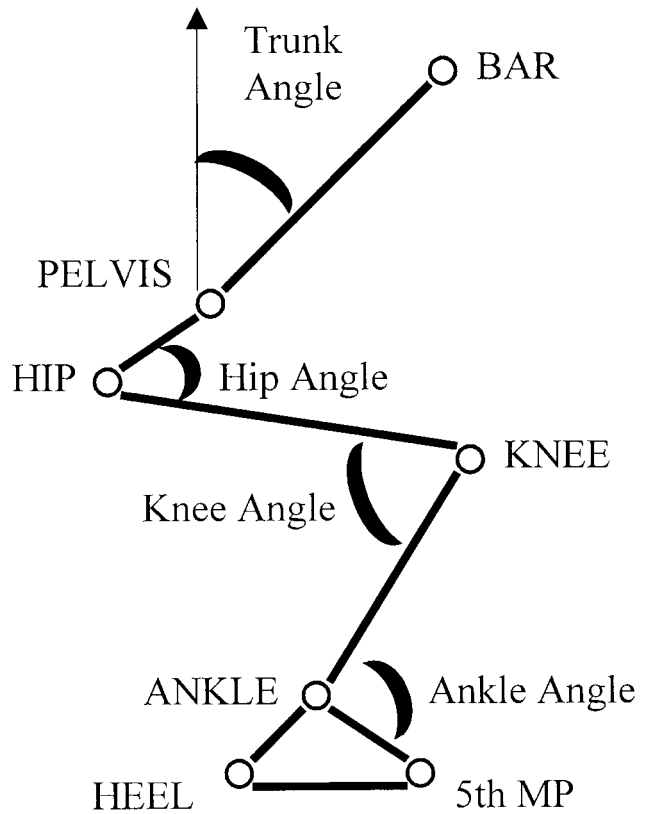

Figure 1. Angle conventions for kinematic analysis.

Data Analysis

The kinematic and force plate data were used to identify 3 positions in each lift: P1, start of descent (determined by first decrease in vertical force from the baseline of combined body and barbell weight); P2, bottom position (determined by maximum knee flexion); and P3, end of ascent (determined by vertical force return to baseline). These 3 positions were used to divide each lift into 2 phases: the downward phase (DP) from P1 to P2, and the upward phase (UP) from P2 to P3. Videographic and force plate data were used to cal-

Table 1. Temporal values (seconds).†

	NWB		WB	
	Mean	*SD*	Mean	*SD*
Overall	3.21	0.77	2.96*	0.65
Downward phase	1.71	0.43	1.60**	0.42
Upward phase	1.50	0.43	1.36**	0.36

* $p < 0.0033$ (0.01/3).
** $p < 0.0167$ (0.05/3).
† NWB = without weight belt; WB = with weight belt.

culate the duration of the 2 phases. Kinematic data were smoothed using a fourth-order, zero-lag Butterworth filter, and the smoothed coordinates were used to calculate angular displacements and velocities of the ankle, knee, hip, and trunk.

EMG signals were full-wave rectified and integrated, with peak EMG value, time-to-peak EMG value, and mean value (mEMG) determined for both phases of each lift. The mEMG values were normalized for each subject based on the maximal peak EMG value (100%) observed during the 2 defined phases of the lift across the trials performed both with and without a weight belt.

Statistical Analyses

A multiple ANOVA test for difference was conducted ($p \leq 0.05$) to determine if a difference in mean EMG values and time-to-peak EMG existed between WB and NWB conditions in either of the 2 phases for any of the 5 muscles.

Descriptive statistics were calculated for range of motion (ROM); maximum and minimum joint angles for the trunk, hip, knee, and ankle; and the mean angular velocity and peak angular velocity for the hip, knee, and ankle for both the DP and UP. Descriptive statistics were also calculated for linear displacements of the bar, hip, knee, and ankle and mean linear velocities and peak velocities of the barbell for both the DP and UP. A dependent *t*-test with a Bonferroni adjustment was used to determine if differences existed between belted and nonbelted conditions for kinematic variables of interest (6).

Results

Timing

Temporal data for both conditions are summarized in Table 1. WB trials were significantly shorter ($p < 0.0167$) than NWB trials in overall time and both the DP and UP of the lift. Trials conducted with a weight belt demonstrated a decrease in durations of the DP by 6.6%, the UP by 9.1%, and overall by 7.7%.

EMG

The *F* values for the interaction between WB and NWB conditions for both mEMG and time-to-peak EMG were not significant during either phase for any of the 5 muscles. Figure 2 displays the mEMG of the 5 muscles tested for both phases as a percent of peak EMG.

Kinematics

No significant differences were detected between WB and NWB conditions for any angular kinematic variables (Tables 2 and 3). However, there was a trend of increased trunk, knee, and ankle ROM in the WB condition. The total distance traveled by the barbell mark-

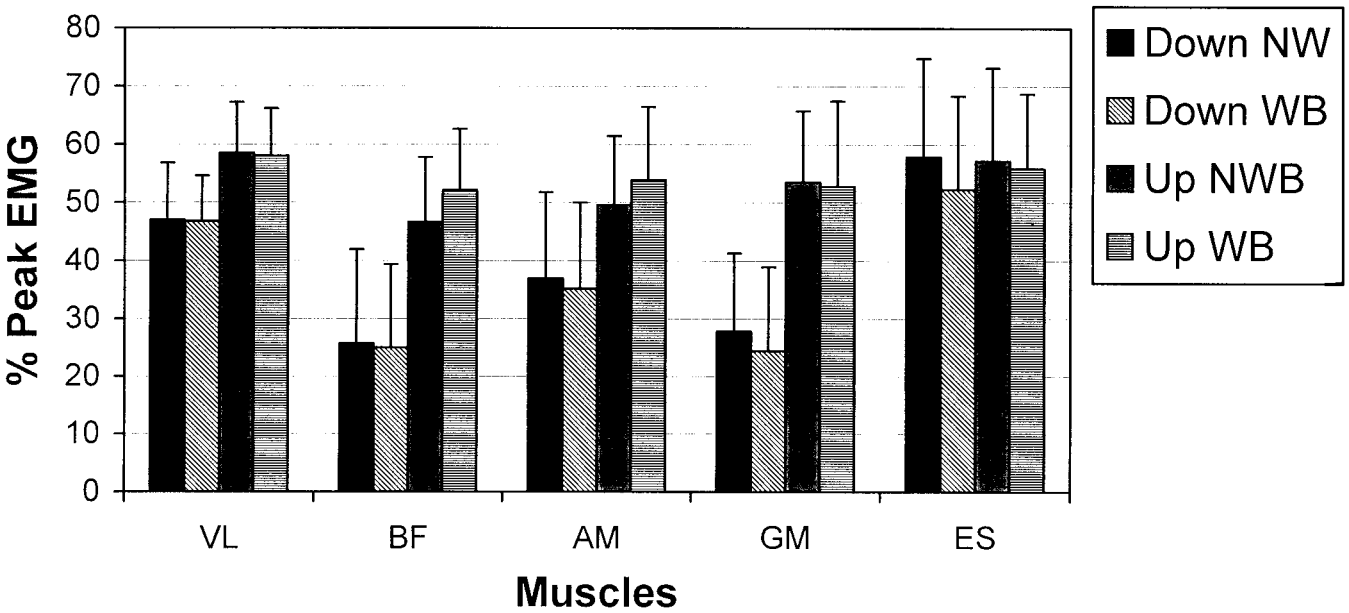

Figure 2. Mean EMG (mEMG) for vastus lateralis (VL), biceps femoris (BF), adductor magnus (AM), gluteus maximus (GM), and erector spinae (ES) represented as a percentage of peak EMG.

Table 2. Joint angular displacements (degrees).*

	NWB		WB	
	Mean	*SD*	Mean	*SD*
Range of motion				
Trunk	23	9	25	11
Hip	70	20	68	18
Knee	78	15	80	14
Ankle	24	4	25	4
Maximum angle				
Trunk	3	7	4	6
Hip	155	10	152	9
Knee	160	6	161	6
Ankle	113	10	115	11
Minimum angle				
Trunk	26	13	29	14
Hip	86	20	84	19
Knee	82	13	80	13
Ankle	89	11	89	10

* NWB = without weight belt; WB = with weight belt.

Table 3. Joint angular velocities (degrees per second).*

	NWB		WB	
	Mean	*SD*	Mean	*SD*
Mean velocity				
Hip				
Down phase	48	12	51	12
Up phase	42	11	44	11
Knee				
Down phase	56	17	62	17
Up phase	48	12	53	13
Ankle				
Down phase	15	5	17	5
Up phase	18	8	21	8
Peak velocity				
Hip				
Down phase	95	32	100	31
Up phase	96	28	100	29
Knee				
Down phase	123	31	136	33
Up phase	139	45	144	44
Ankle				
Down phase	52	23	56	18
Up phase	65	16	74	19

* NWB = without weight belt; WB = with weight belt.

er in both the anterior-posterior (A-P) and vertical directions was significantly different ($p < 0.0083$) between conditions (Table 4). The absolute displacement increased by 26.0% anteriorly and 5.4% vertically in the WB condition when compared with the NWB conditions. No other joint marker linear displacements differed significantly between conditions.

Table 4. Linear displacements (meters).†

	NWB		WB	
	Mean	*SD*	Mean	*SD*
Total displacement				
Barbell				
X	0.077	0.052	0.097*	0.047
Y	0.156	0.136	0.544*	0.136
Hip				
X	0.119	0.047	0.108	0.034
Y	0.310	0.077	0.316	0.070
Knee				
X	0.179	0.037	0.182	0.033
Y	0.092	0.024	0.095	0.019

* $p < 0.0083$ (0.05/6).
† NWB = without weight belt; WB = with weight belt.

Table 5. Linear velocities (meters per second).†

	NWB		WB	
	Mean	*SD*	Mean	*SD*
Mean velocity (barbell)				
X				
Down	0.044	0.021	0.062*	0.028
Up	0.052	0.028	0.073*	0.033
Y				
Down	0.309	0.072	0.350**	0.079
Up	0.361	0.099	0.417*	0.115
Peak velocity (barbell)				
X				
Down	0.095	0.034	0.100	0.033
Up	0.086	0.034	0.093	0.014
Y				
Down	0.640	0.182	0.634	0.184
Up	0.745	0.243	0.839	0.236

* $p < 0.0025$ (0.01/4).
** $p < 0.00025$ (0.001/4).
† NWB = without weight belt; WB = with weight belt.

Both the A-P and vertical velocity of the barbell were significantly greater ($p < 0.0025$) between the NWB and WB conditions in both the DP and UP (Table 5). Between conditions the mean vertical velocity was greater for the WB condition by 13.3% in the DP and 15.5% in the UP, whereas the mean A-P velocity increased by 40.9 and 40.4% in the DP and UP, respectively.

Discussion

This study was designed to investigate the effects of weight-belt use during the squat lift performed at heavy loads (90% of 1RM) in lifters using self-selected speeds and technique. This load was chosen based on findings by Lander et al. (3), which demonstrated greater changes in variables studied at 90% of 1RM than at 80 or 70%. Experienced weight-trained individuals were used to ensure that similar squat technique was used between conditions under heavy loads. Subjects demonstrated a mean ± *SD* 1RM without the use of a weight belt of 156.0 ± 20.4 kg, or 1.8 ± 0.2 times body weight. The lack of significant difference between WB and NWB conditions for either angular or linear joint kinematic variables suggests similar technique regardless of condition. However, the trend of increased trunk, knee, and ankle ROM in the WB conditions may indicate subtle changes in the ROM of each joint that were not detected with the measurement apparatus or statistical tools used.

The observation of greater anterior displacement of the barbell marker in the WB condition is consistent with that of Lander et al. (3), who reported that using a weight belt shifted the lifter's center of mass anteriorly to a greater extent than when not using a weight belt. The trend of increased ROM at the knee, ankle, and trunk may partially explain the increase in overall bar depth and anterior displacement that was detected. Small changes in ROM of the knee, ankle, and trunk may not be significant individually but may have a cumulative effect of increasing both the distance that the bar is lowered and the amount of anterior displacement at the end of the DP.

For safety reasons, subjects were allowed to self-select lifting speed and performed squats with a weight belt faster overall, as well as in both the DP and UP of the lift. This agrees with previous studies of squats performed with weight belts (2, 3). All subjects in this study reported a feeling of greater security and trunk stability when wearing a weight belt. This may partially explain the decrease in duration of the lift when wearing a weight belt.

The vertical and A-P velocities in both phases were greater in the WB condition than the NWB condition (Table 5). A nonsignificant trend of increased angular velocity and peak angular velocity of the hip, knee, and ankle also existed in both the DP and UP in the WB condition when compared with the NWB condition. These results are a logical consequence of the similar ROM between conditions and the decreased duration of the lift phases.

EMG data showed no significant differences for mEMG or time-to-peak EMG between WB and NWB conditions for any of the muscles recorded during either the DP or UP. These results are in disagreement with a study by Lander et al. (2), which reported an increase in VL and BF mEMG activity during the UP of the squat exercise with the use of a weight belt. This discrepancy is explainable by differences in experimental design. Their protocol required subjects to perform multiple repetitions of the squat exercise at 80% of 1RM. Many of the mEMG increases in the UP in that study occurred during later repetitions, and their results therefore may have been influenced by fatigue, which has been widely reported to increase mEMG.

The results of this study also demonstrated that mEMG activity of the ES is unaffected by weight-belt use. This finding is consistent with previous studies (2, 3, 5), which reported no significant difference in mEMG of the ES between belted and nonbelted conditions while performing the squat movements.

Although it was not significant, there was a trend toward increased BF and AM myoelectric activity in the upward phase and decreased GM and ES activity in the downward phase of the lift in the WB condition (Figure 2). The decrease in mEMG for the GM and ES in the WB condition may partially account for the shorter DP duration observed as the subjects descended with decreased eccentric activation of those muscles. This change may also be an indicator of decreased lower-back and hip-extensor recruitment due to partial support of the load by the weight belt. The increase in mEMG for the BF and AM may indicate a greater reliance on hip extensors and adductors during the UP with a weight belt. This may account for the lack of difference in mEMG of the VL between conditions due to a decreased reliance upon knee extensors during ascent.

The increase in anterior displacement of the barbell in the bottom position decreases the moment arm and the torque generated about the knee while increasing the moment arm and the torque generated about the hip. Although less force would be needed to counter the torque at the knee, a greater countertorque would need to be generated by the back and hip extensors to compensate for the increased moment arm at the hip. However, there was a trend toward decreased myoelectric activity of the GM and ES. Perhaps the weight belt served to partially support the load even with the increased barbell anterior displacement and subsequent torque generated about the hip. This difference in bar position induced by weight-belt use may indicate a subtle change in squat technique that results in different muscle recruitment with the GM and ES recruited less and the BF and AM recruited more. Further study is needed to resolve these issues.

The lack of significant differences in EMG between conditions may be attributed to the lack of large differences in mEMG and the high *SD*s in the mEMG data. This may be due in part to subject variability in squat techniques related to their different sport backgrounds and training regimens. The high-bar squat technique employed by weightlifters has been reported

to differ from the low-bar squat technique employed by powerlifters in both joint angles and joint moments for the hip and knee (7). Differences in squat technique such as those exhibited between low-bar and high-bar squatters may have existed between subjects in this study, and therefore may be partially responsible for the high variability in EMG.

The use of a weight belt during the squat exercise does not appear to affect the myoelectric activity of the leg or back extensor muscles. Similarly, individual angular and linear joint kinematics were not significantly changed with the use of a weight belt. However, the total distance that the barbell traveled anteriorly and vertically was greater and the lifts were performed at a faster rate during both phases of the lift when a weight belt was worn. This suggests that weight-belt use during the squat exercise can alter the speed of the lift, the path the barbell travels, and the position of the barbell at the end of the downward phase without significantly altering the overall muscle activity.

Practical Applications

One of the best ways to improve performance and reduce the risk of injury during heavy lifts is to maintain proper lifting technique. Comprehensive assessment of lifting technique must consider anatomical, mechanical, and physiological aspects of the lift.

This study showed that at near-maximal loads, lifters using a weight belt maintain a horizontal barbell position closer to the knee and farther from the hip at the bottom of the lift. This bar position would result in greater torques about the hip and lesser torques at the knee. In addition, the fact that belted lifts showed more vertical bar displacement over a shorter time interval, when compared to non-belted lifts, suggests that lifters wearing a weight belt perform more mechanical work and generate more power. Further study is needed to confirm these observations and determine whether these kinetic differences are of practical relevance.

The experienced lifters in this study showed biomechanical differences between the WB and NWB conditions. From a performance perspective, the faster lifting associated with weight-belt use may provide a slight positive benefit in terms of increased power output. In contrast, an argument can be made that the slower and more controlled movements without belt use may be preferred in terms of reducing risk of injury. Use of this study's results should be made with care. Interpretation and application of the data may differ depending on the target population (e.g., athletes in general, powerlifters) and training goals (e.g., general fitness, increased power output).

As is the case in most near-maximal performance situations, the line between optimum performance and injury potential is a fine one. In that light, questions related to the recommended technique for near-maximal lifting will not be fully resolved until study is made of the joint- and tissue-loading profiles associated with such lifts.

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